

Research Fellows Workshop

Research data analysis with generative AI

From file analysis to charts, reports and slides

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Which tasks, and what they expect from it

02 File analysis: tables, figures, reports p.13

Read Excel, compute, generate outputs, revise

03 Building reusable analysis tools p.28

Describe the features you need, get a web tool

04 Comparative analysis and synthesis reports p.31

Vary methods and assumptions, compare, write up

05 Driving existing desktop software p.35

Linking Excel, OriginPro and Word

06 Generating slides and applying design p.38

Build a template from a reference, revise an existing PPT

01

Where and how much do researchers use AI

"AI for Science 2026, Part I" researcher survey

01 Survey scope and respondents

02 Use by task, and leading tools

03 Hopes and concerns about AI

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Survey overview

Valid responses

10,480 respondents

Surveyed 2-30 March 2026

Coverage

117

countries and economies, 12 research fields

Academia

74 %

plus hospitals (11%), government (5%), industry (4%)

Weighting

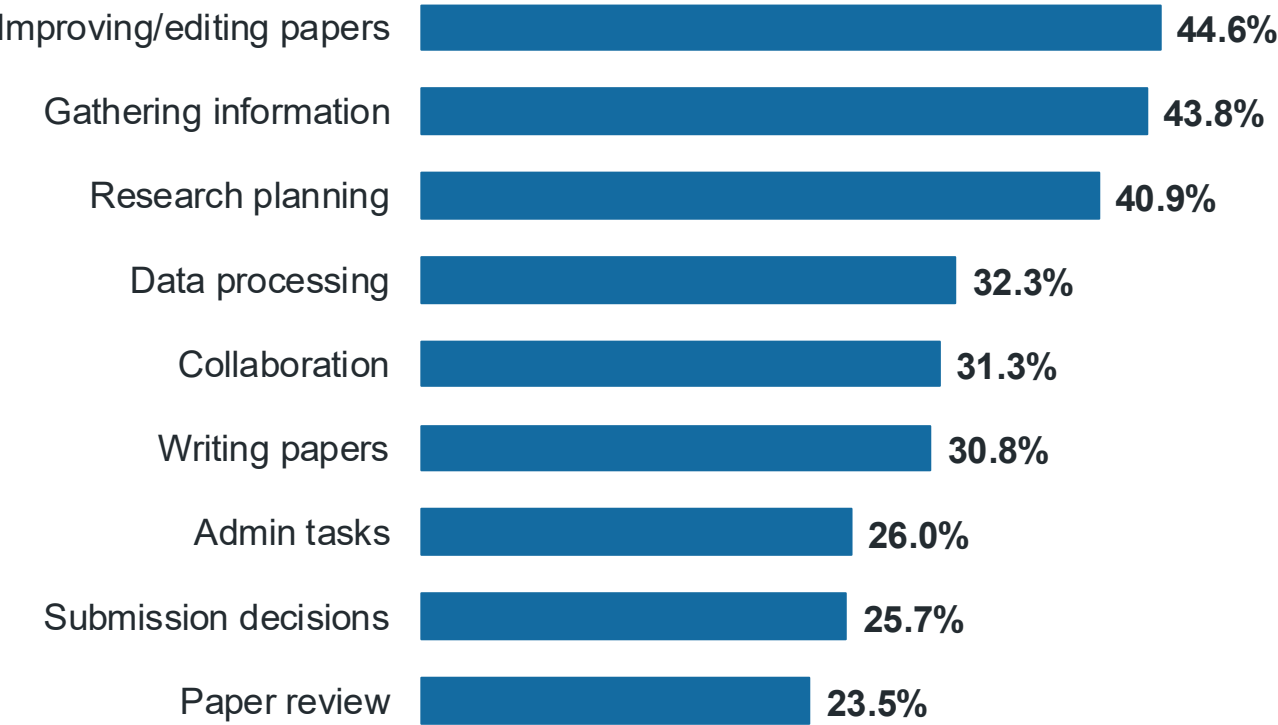
Weighted by 2024 national publication output

Scope

Responses centre on researchers published in international journals
Not representative of agricultural research or of all researchers

AI use by research task

Frequent use = always + mostly



44.6%

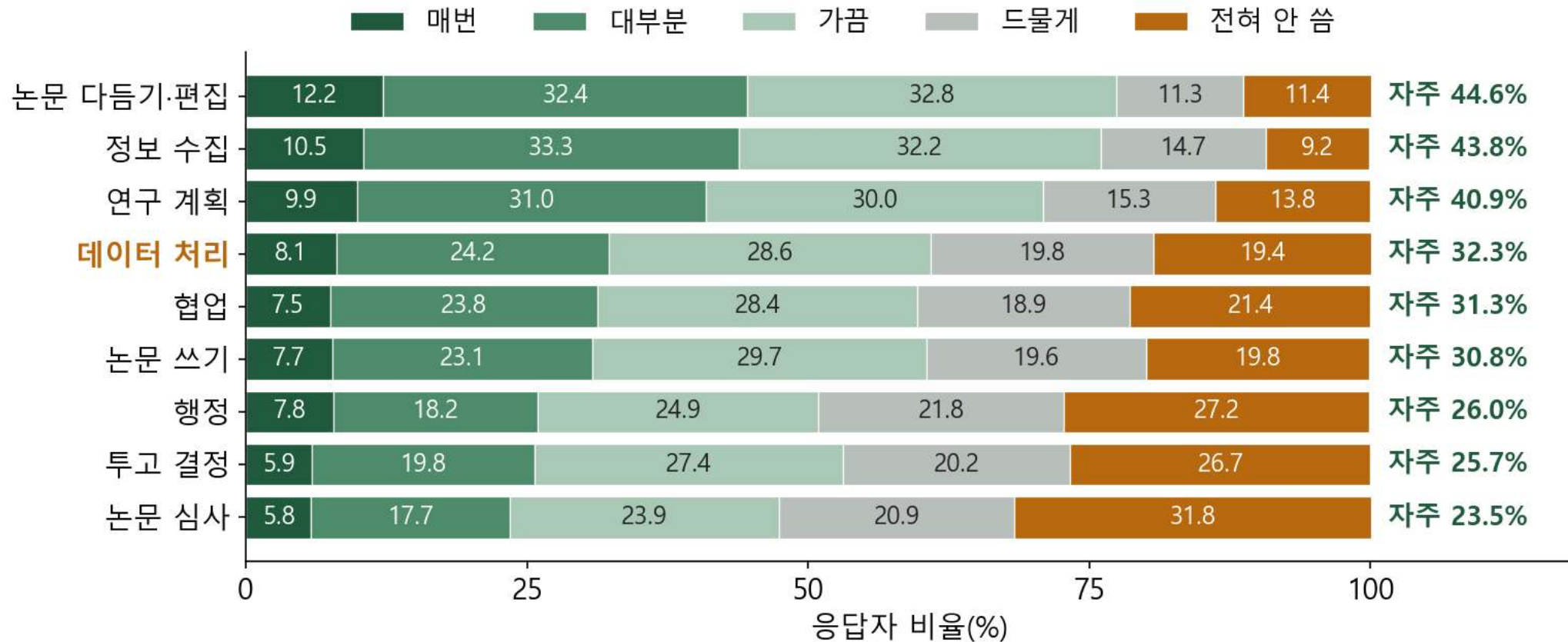
always or mostly use AI
to improve and edit papers

- Review and submission decisions
show low rates of frequent use

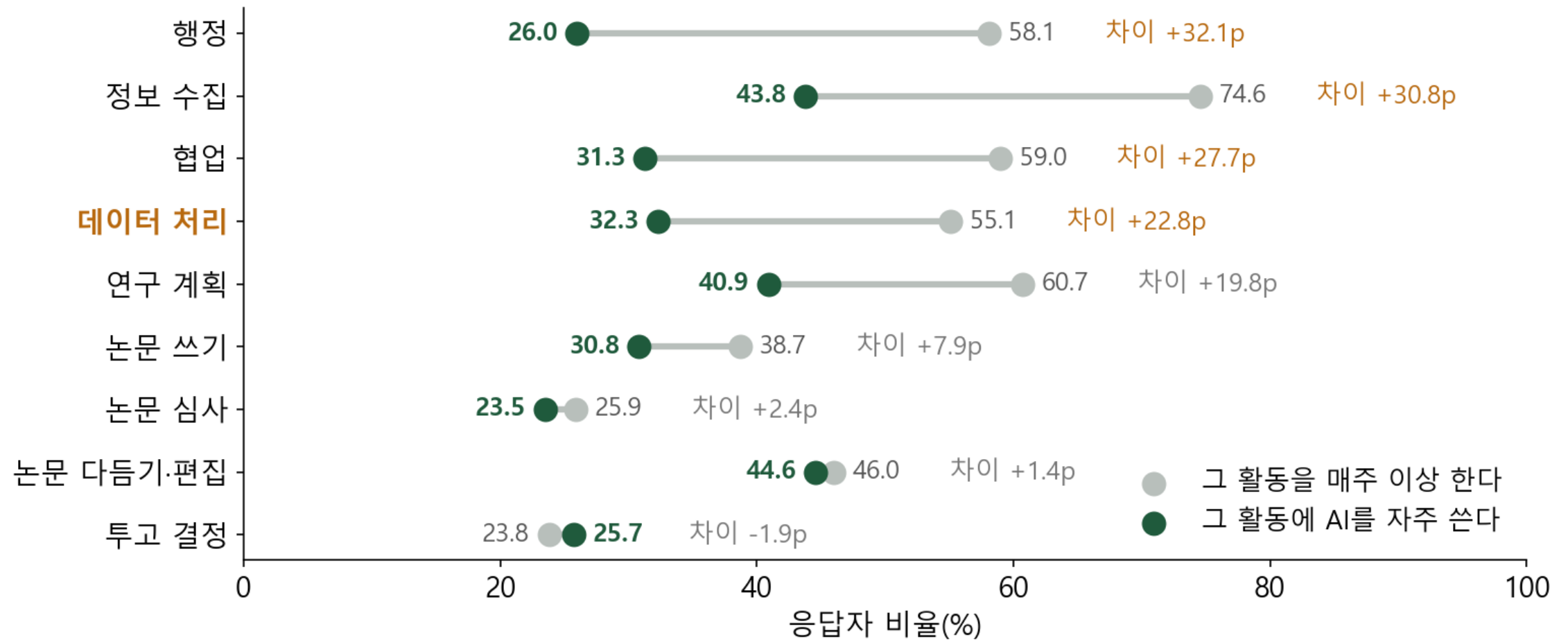
AI uptake lags where judgement is required

AI use by research task, 5-point responses

"Never" responses run high where decisions are involved



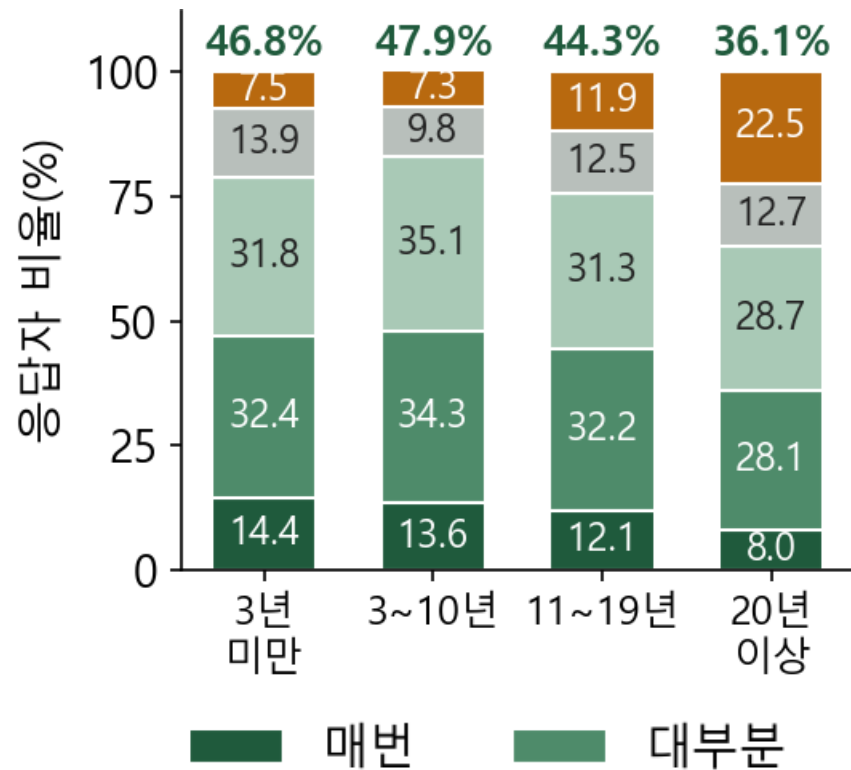
The gap between what researchers do and where AI has entered



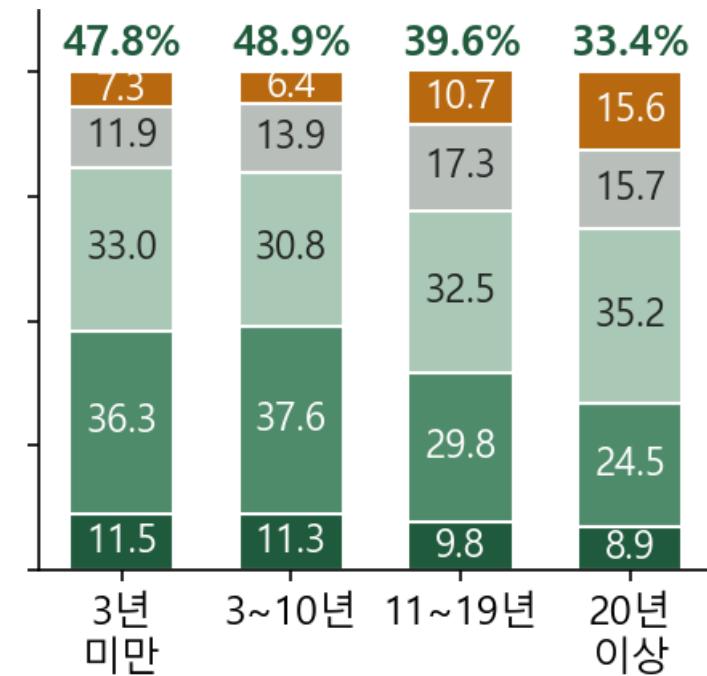
The longer the career, the less the use

Figures on top = share answering 'always + mostly'

Polishing and editing papers

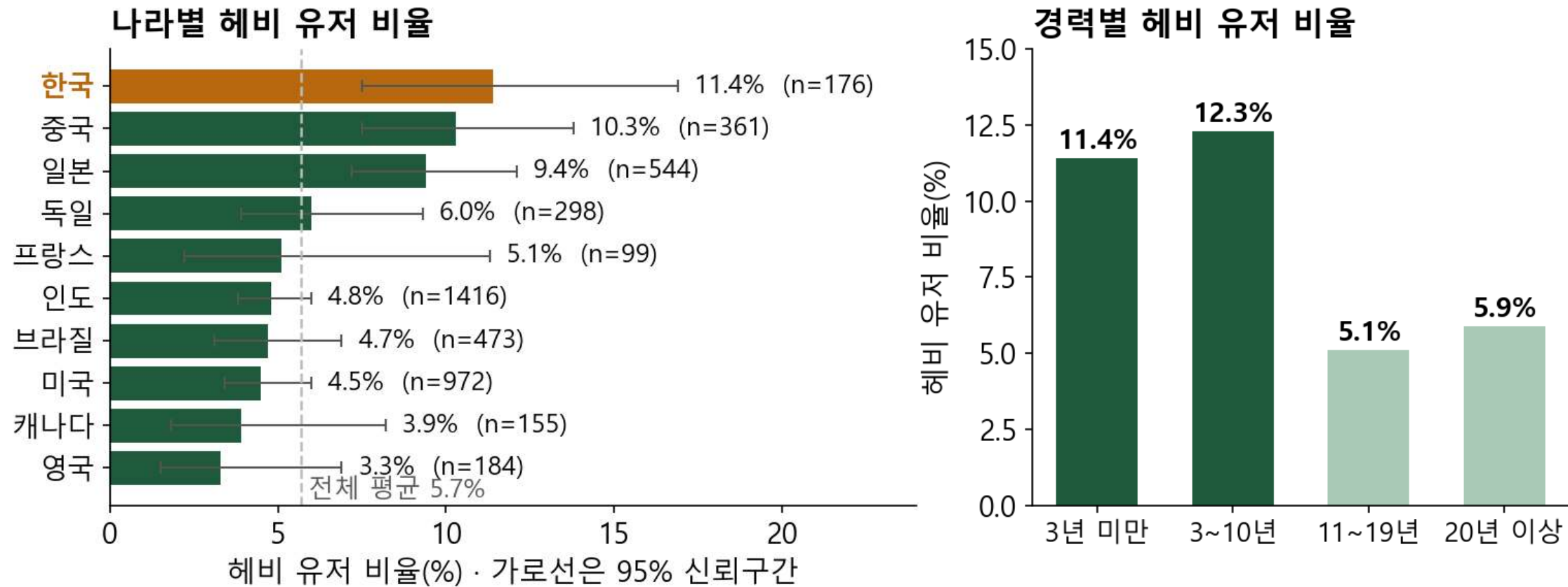


Gathering information



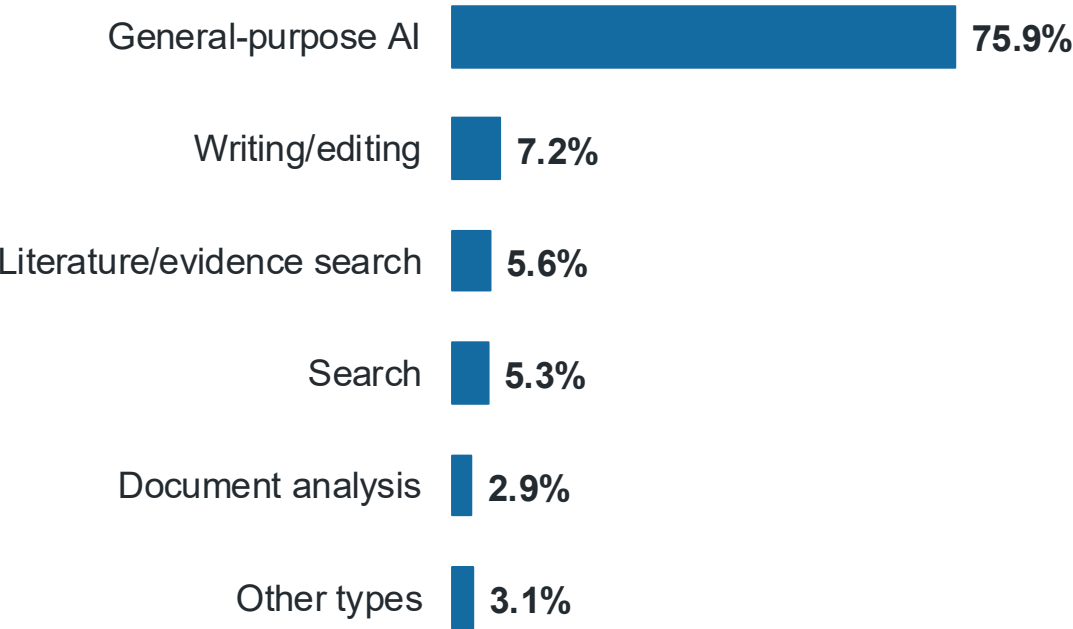
By country, and Korea

Heavy users: those relying mostly or entirely on AI for literature search

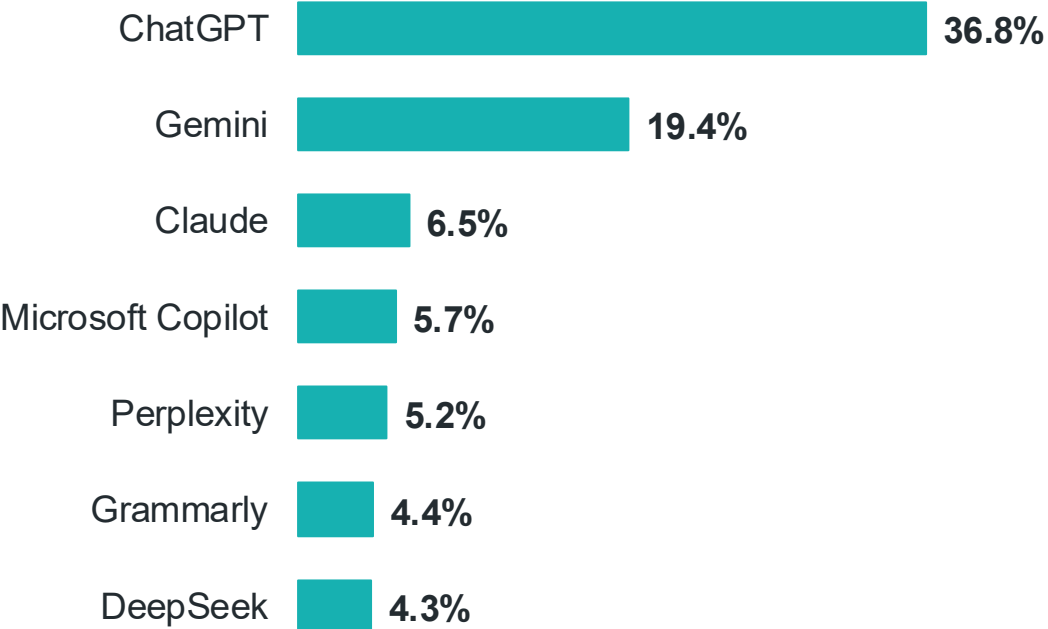


Leading AI tools named by researchers

① Share of mentions by tool type



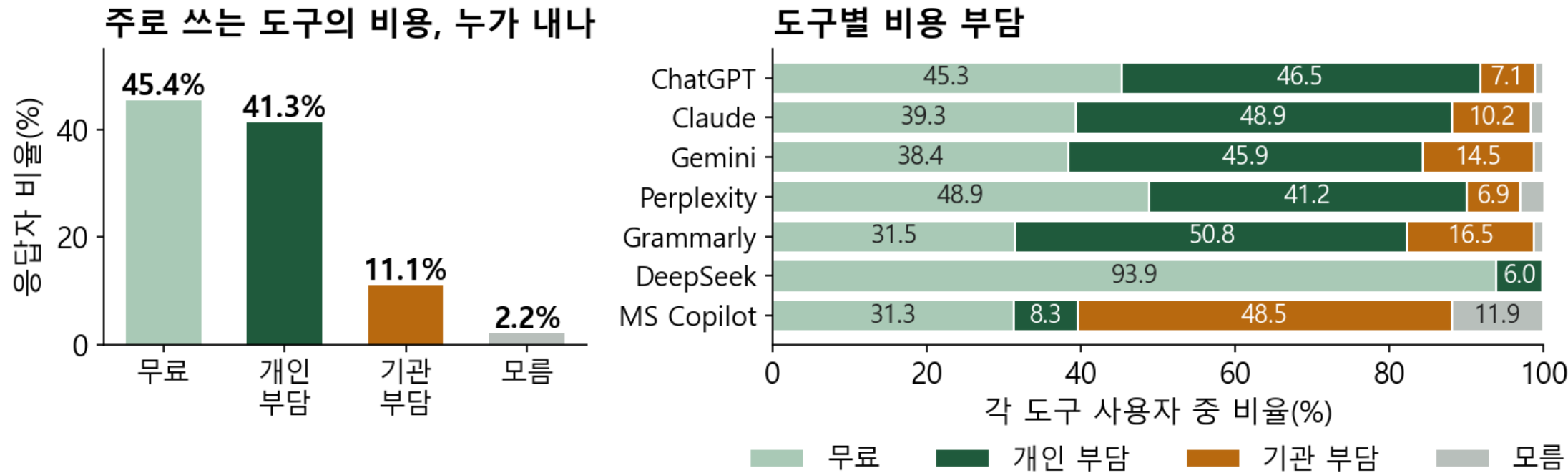
② Share of mentions, top 7 tools



Up to 3 tools each · based on 23,842 mentions

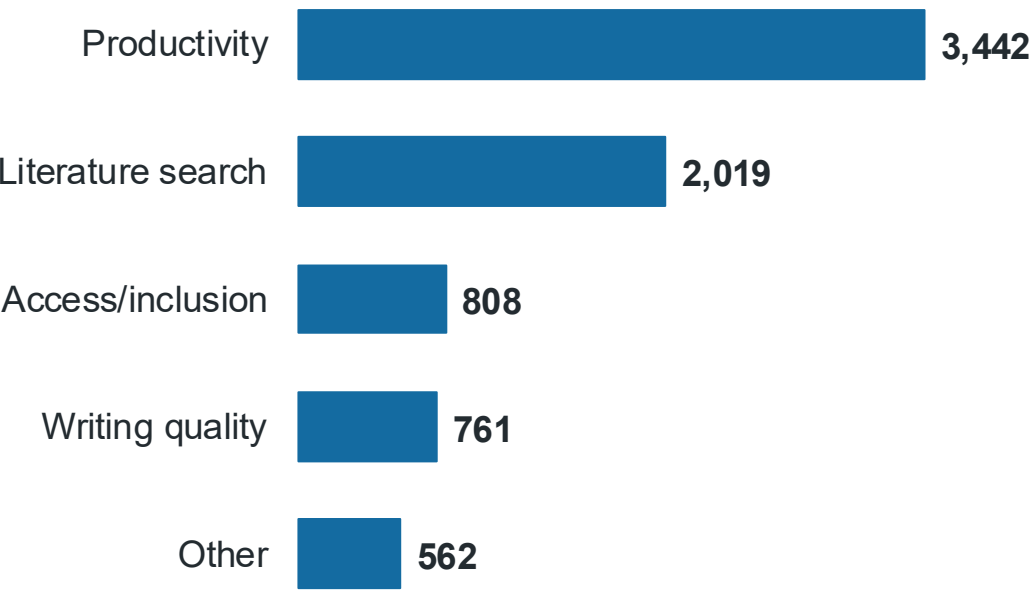
Who pays

Nine in ten use free tools or pay out of pocket

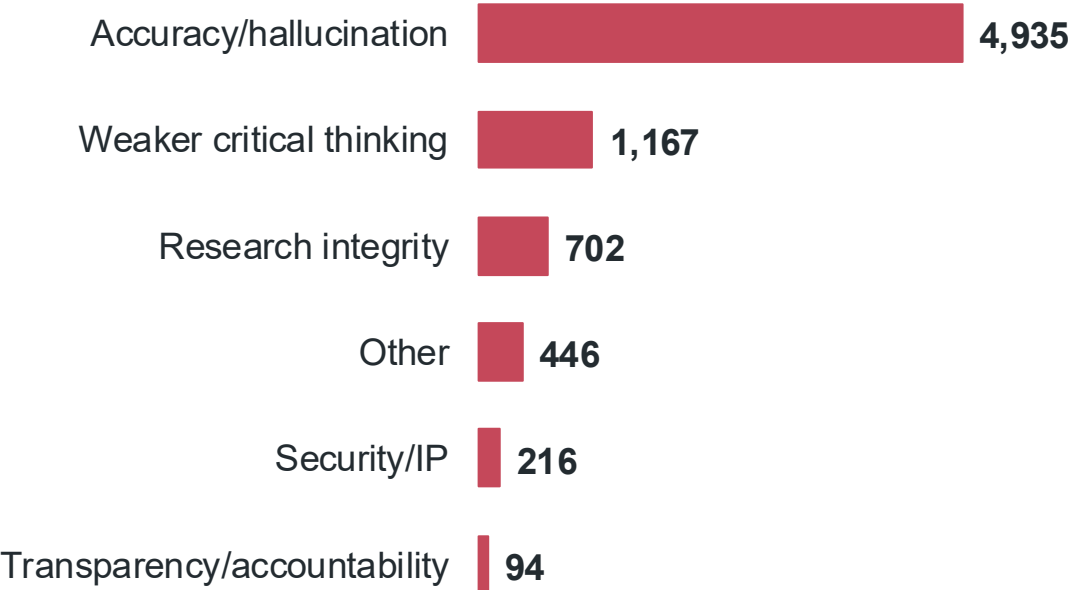


Hopes and concerns about AI

① 7,592 free-text responses on hopes



② 7,560 free-text responses on concerns



Productivity tops hopes; accuracy and hallucination top concerns

Hallucination: presenting unfounded content as fact

02

From research files to tables, figures and reports

- 01 App setup and first Excel analysis
- 02 Refining figures, writing a Word report
- 03 Expanding data scope and methods

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- 01 How researchers use AI today
- 02 File analysis: tables, figures, reports**
- 03 Building reusable analysis tools
- 04 Comparative analysis and synthesis reports
- 05 Driving existing desktop software
- 06 Generating slides and applying design

AI and skills for working with research files

Claude Code·Codex

Reads files in the working folder,
writes and runs code,
and builds tables, figures, reports

Skill

Repeatable procedures with output and
check rules

R

Language for computation
and graphs

Ask and verify

Describe files and desired output, check results, request revisions

Worked example: 2023 Anseong soybean rocket-trap data

Installing the Claude app and starting Code

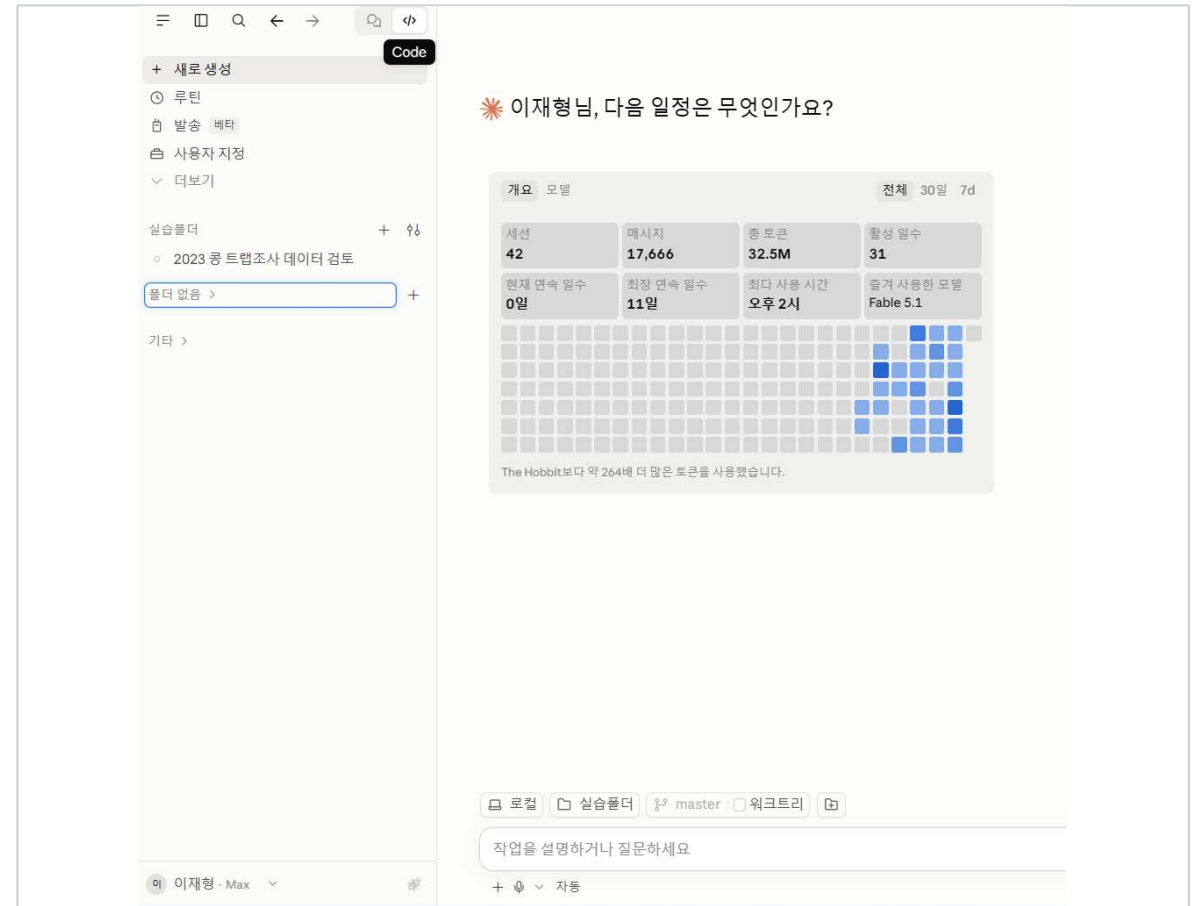
1 Install and sign in to the Claude app

2 Select the Code tab

3 Choose local work and a folder

Install: claude.ai/download

Screens as of the session · menus vary by version



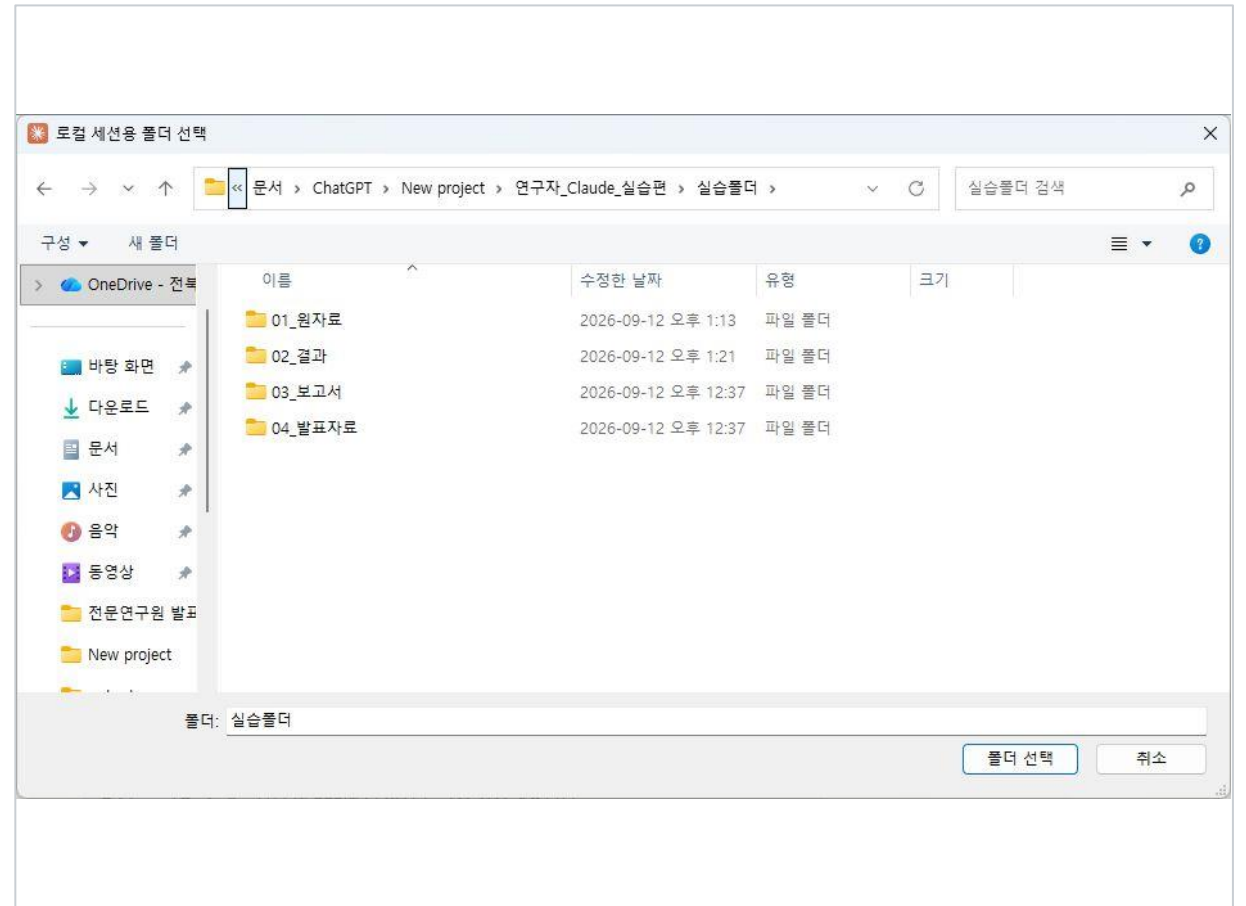
Choosing a working folder

1 Create a practice folder

2 Select it in Code

3 Put the Excel file in the folder

e.g. 01_원자료/2023_콩_트랩조사.xlsx



Input Excel file and calculation conditions

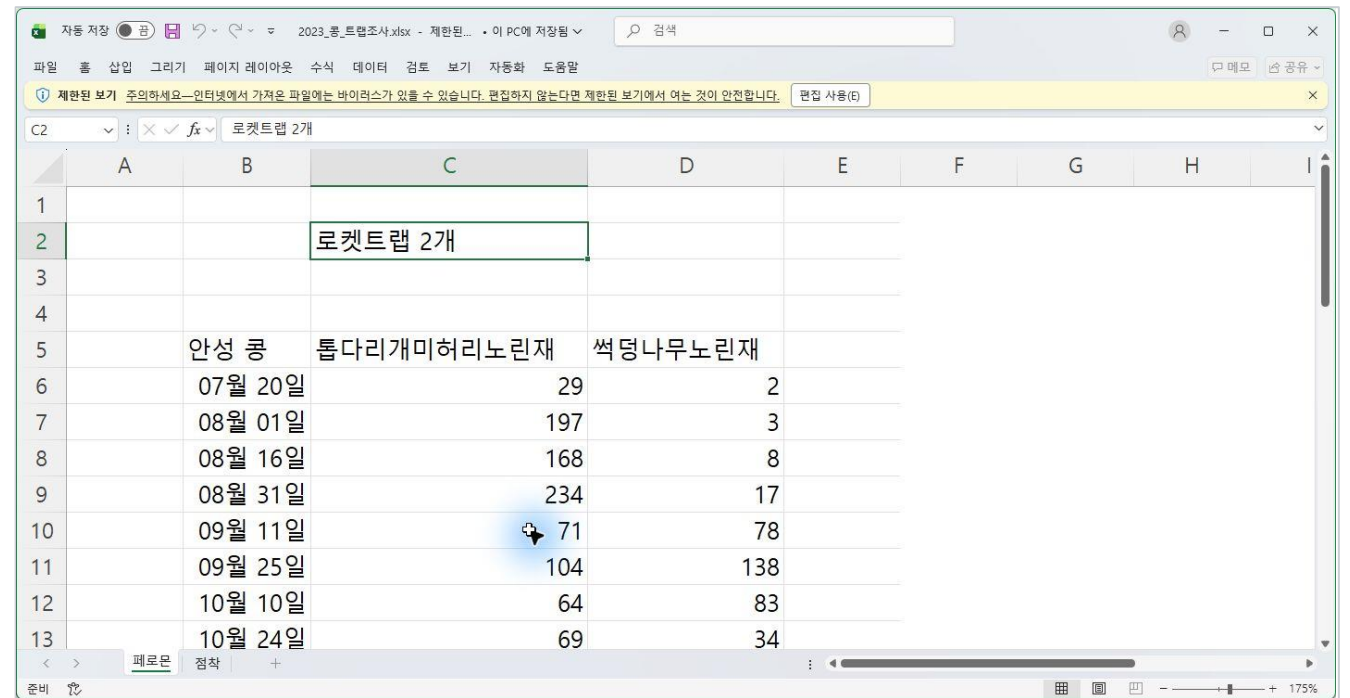
2023_콩_트랩조사.xlsx

- Anseong soybean, B5:D13 on the Pheromone sheet
- Trap catch = sum of 2 rocket traps
- 8 survey dates × 2 species = 16 values

Calculation to request

$$29 \div 2 = 14.5$$

- Per-trap mean, first survey date
- Unit: insects/trap



	A	B	C	D	E	F	G	H	I
1									
2			로켓트랩 2개						
3									
4									
5		안성 콩	톱다리개미허리노린재	씩덩나무노린재					
6		07월 20일	29	2					
7		08월 01일	197	3					
8		08월 16일	168	8					
9		08월 31일	234	17					
10		09월 11일	71	78					
11		09월 25일	104	138					
12		10월 10일	64	83					
13		10월 24일	69	34					

Setting the scope of the first prompt

Prompt to enter

Read 2023_콩_트랩조사.xlsx and describe the sheets and table structure.

The target is Anseong soybean, B5:D13 on the Pheromone sheet.

Check survey dates, trap catch by pest and notes on trap number, and ask about anything unclear.

Do not start the analysis yet, and leave the original file unchanged.

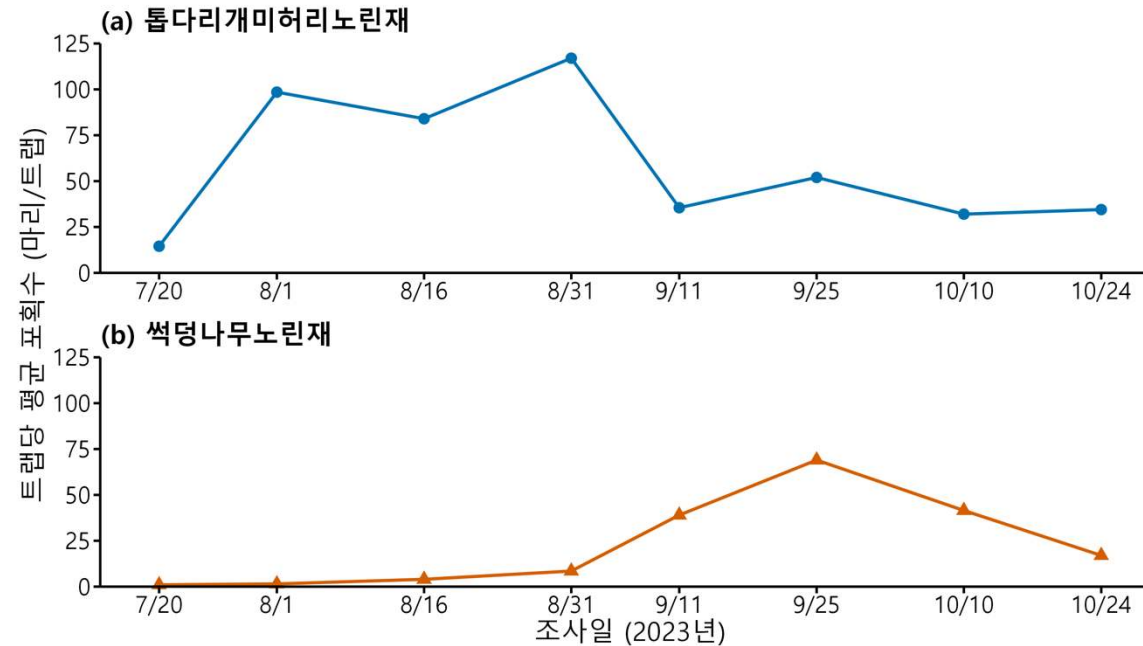
Finding missing values and outlier candidates

- Check missing values, duplicate dates, text-formatted numbers, negatives
- Record and check cells with sudden jumps
- Decide on corrections after checking the field notebook

Clarifying question from the actual AI response · excerpt

1. C7 and D10 differ greatly from earlier values. Please check the field notebook

Finding what to fix in the first result



- First result keeps colours and a stacked layout
- The reference figure's B&W lines and legend were missing from the prompt

Revising the R graph with a reference figure

① What to keep

- Each of the 16 trap catches divided by 2
- Actual survey dates and the two pests distinguished

② Reference file and tools

- Reference figure, journal-style graph
(full path in the notes)
- r-stat-graphs skill
R runtime

③ Prompt summary

- Show both species in one panel.
- Bean bug: black solid line, filled circles
- Stink bug: black dashed line, open triangles
- Legend on top, white background, no gridlines
- Save at 90×64 mm and keep the aspect ratio in PPT

Full prompt and file paths are in the speaker notes

Applying the skill

Check available skills



Read instructions



Request application

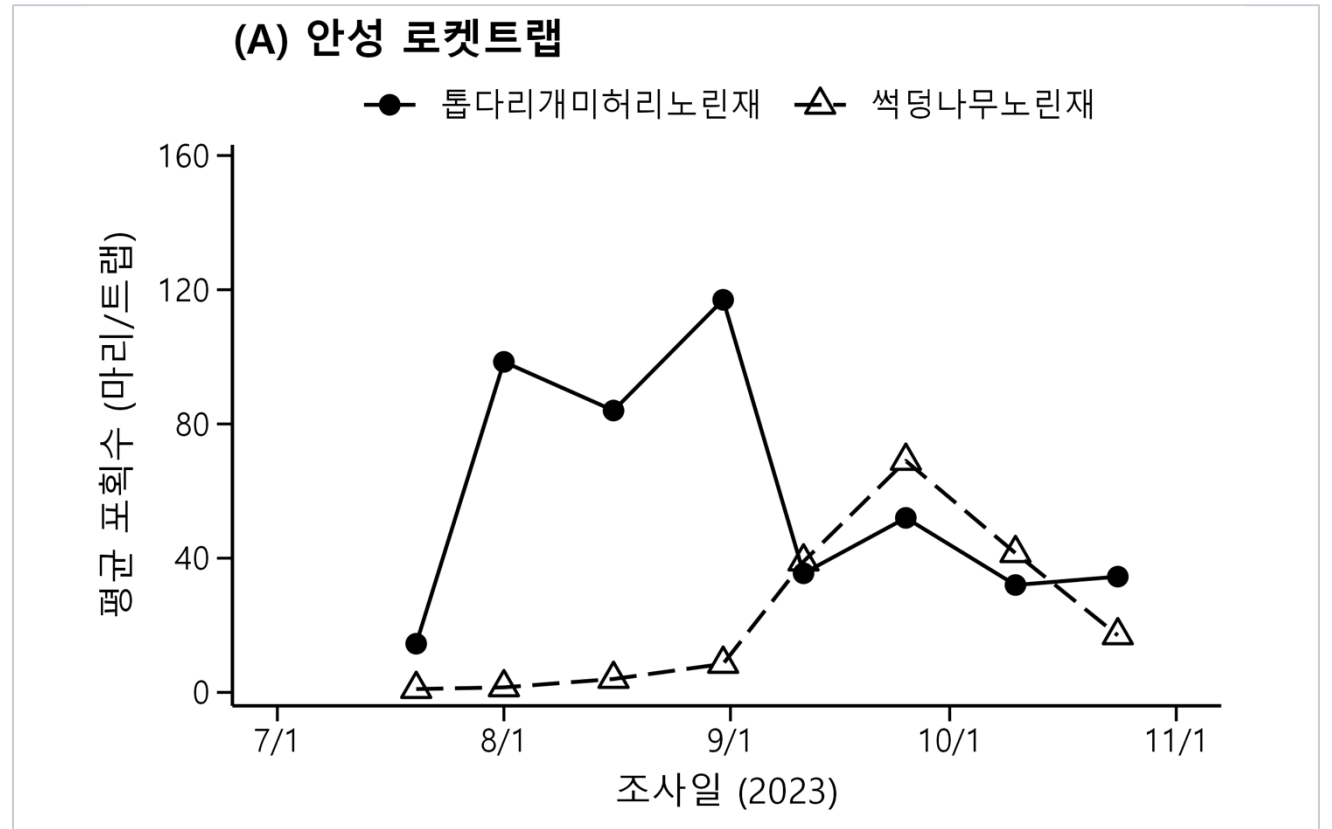


Compare results

Checking the revised journal-style graph

- Both species' trends in one panel
- Solid/filled vs dashed/open markers
- Points between monthly ticks are actual survey dates
- Check axis units and the 16 values against Excel

Fig. 1. Mean trap catch by survey date (insects/trap)



Compiling tables and figures into a Word report

- Compile the results from that Excel file into a report

4. 결과 설명

4.1 톱다리개미허리노린재

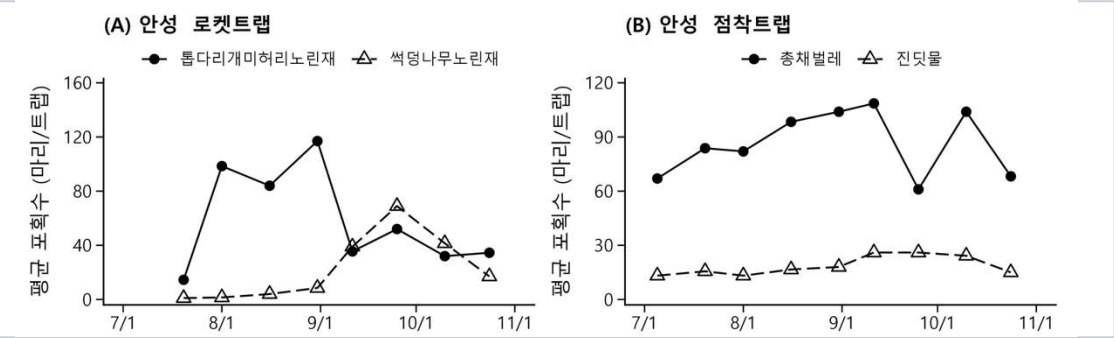
트랩당 평균 포획수는 7 월 20 일 14.5 마리/트랩, 8 월 1 일 98.5 마리/트랩이었다. [표 1: 7/20-8/1 행 | 원본 C6-C7
→ 정리표 E6-E7]

8 월 16 일 84.0 마리/트랩이었고, 조사 기간 최댓값은 8 월 31 일 117.0 마리/트랩이었다. 최댓값과 같은
값을 보인 다른 조사일은 없었다. [표 1: 8/16-8/31 행 | 원본 C8-C9 → 정리표 E8-E9 | 최댓값!B5:I5(최댓값 조사일 수 1)]

9 월 11 일부터 10 월 24 일까지 4 회 조사값은 32.0~52.0 마리/트랩 범위였다. [표 1: 9/11~10/24 행 | 원본
C10:C13 → 정리표 E10:E13]

Extending scope: sticky traps and regional comparison

① Trap catch by date · Anseong



② Peak catch day among survey dates

Trap	Pest	Peak day	insects/trap
Rocket	Bean bug	8/31	117.0
Rocket	Stink bug	9/25	69.0
Sticky	Thrips	9/11	108.6
Sticky	Aphids	9/11·9/25	26.0

③ Report sentences explaining the results

- 로켓트랩의 톱다리개미허리노린재는 안성 8월 31일 117.0마리/트랩, 서산 9월 1일 147.0마리/트랩으로 조사일 중 가장 높았음. 진안에서는 9월 24일 33.5마리/트랩으로 최대 관측값이 나타났음. 썩덩나무노린재의 최대 관측일은 각각 9월 25일, 10월 11일, 10월 6일로, 세 지역 모두 톱다리개미허리노린재보다 늦었음. 이 조사에서는 두 종을 구분하여 예찰을 지속할 필요가 있음을 보여줌.

Analysis method and output conditions

2025 organic upland soil survey.xlsx

- Years under organic management and GRSP values
- Soil chemistry linked to the same samples

유기재배기간	EE-GRSP(mg/ml)	TE-GRSP(mg/ml)
1	0.1529	0.4016
15	0.1025	0.3018
8	0.1052	0.3088
5	0.1666	0.4714
11	0.1878	0.4888
6	0.1099	0.2690

pH	EC	유기물
5.74	0.93	33.54
7.40	0.30	29.23
7.39	0.62	27.65
6.60	0.86	45.11
6.67	2.70	43.65
7.00	0.30	17.85

Analysis conditions in the prompt

- ① Descriptive statistics for 52 samples
- ② Correlation analysis on specified variables
- ③ Include multiple-testing corrected values
- ④ Save as publication-style table and scatter plots

Publication-ready outputs

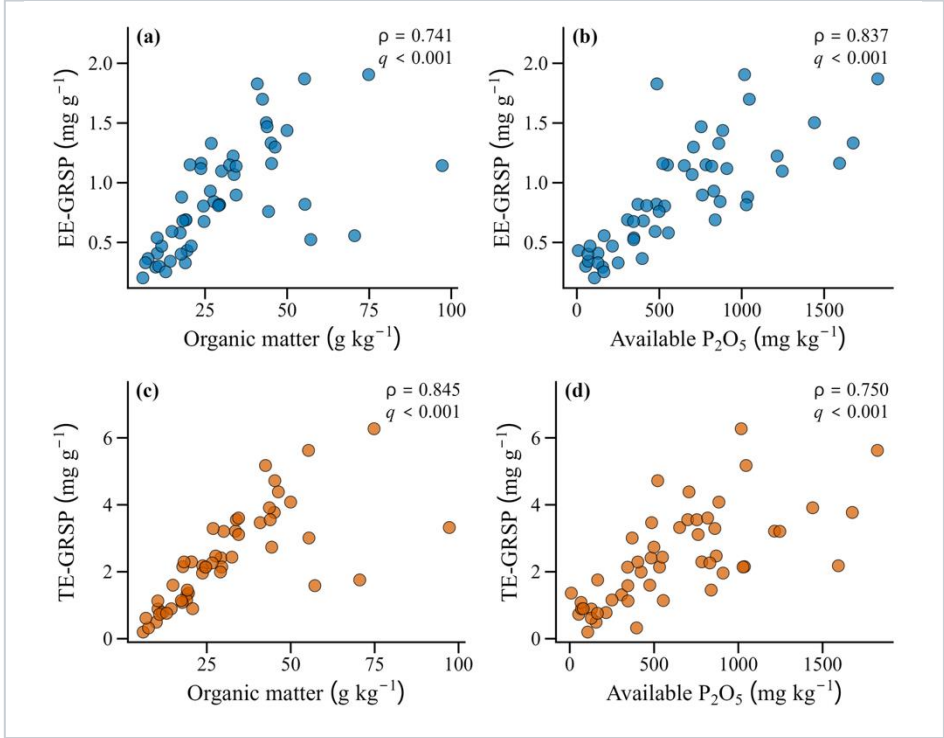
① Descriptive statistics · 52 samples

Table 1. Descriptive statistics of organic cultivation duration, GRSP fractions, and soil chemical properties in soil samples from 52 selected certified organic upland fields.

Variable	Unit	Median	Q1	Q3	Min–Max
Organic cultivation duration	yr	8.0	4.75	15.25	1–27
EE-GRSP	mg g ⁻¹	0.817	0.510	1.153	0.203–1.905
TE-GRSP	mg g ⁻¹	2.166	1.154	3.299	0.199–6.271
pH	1:5 H ₂ O	7.09	6.16	7.51	4.99–8.19
EC	dS m ⁻¹	0.430	0.298	0.718	0.120–2.700
Organic matter	g kg ⁻¹	26.75	17.81	42.78	6.05–97.25
Available P ₂ O ₅	mg kg ⁻¹	526.9	293.0	862.6	9.0–1,825.1
Exchangeable K	cmol _c kg ⁻¹	0.670	0.338	1.388	0.070–5.160
Exchangeable Ca	cmol _c kg ⁻¹	8.23	5.34	10.33	1.03–27.76
Exchangeable Mg	cmol _c kg ⁻¹	1.95	1.21	2.95	0.47–8.57

Q1 and Q3, first and third quartiles; EE-GRSP, easily extractable glomalin-related soil protein; TE-GRSP, total extractable glomalin-related soil protein; EC, electrical conductivity. GRSP values are bovine serum albumin (BSA)-equivalent protein on an air-dried-soil basis, and organic matter was calculated as total C × 1.724.

② Correlation scatter plots



One dot = one sample · ρ : Spearman correlation
 q : p value corrected by BH-FDR

A prompt format you can reuse

① Example folder structure

research_analysis/

raw_data.xlsx

analysis_rules.md

results/

Folder and file names are illustrative

- raw_data: working
- ~~analysis~~ analysis_rules: units, formulas, comparisons
- results: where new outputs are saved

② What to include in the prompt

- 1 Target file and sheet - which data to read
- 2 Items and rules - what to calculate
- 3 Output format - table, figure or document
- 4 Save location - which folder

e.g. "Check the trap number and how catches are aggregated on the Pheromone sheet, then make a mean-per-trap graph in the results folder."

After the first result, keep iterating: "change the date format", "add a table too"

03

Building reusable analysis tools

Describe the features → get a screen for settings, calculation, saving

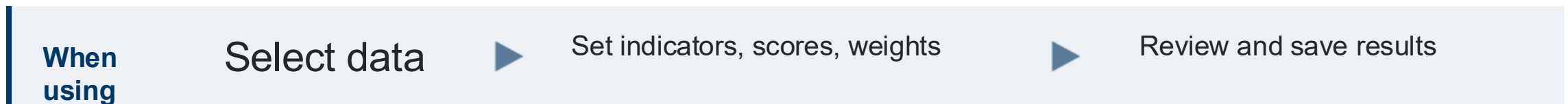
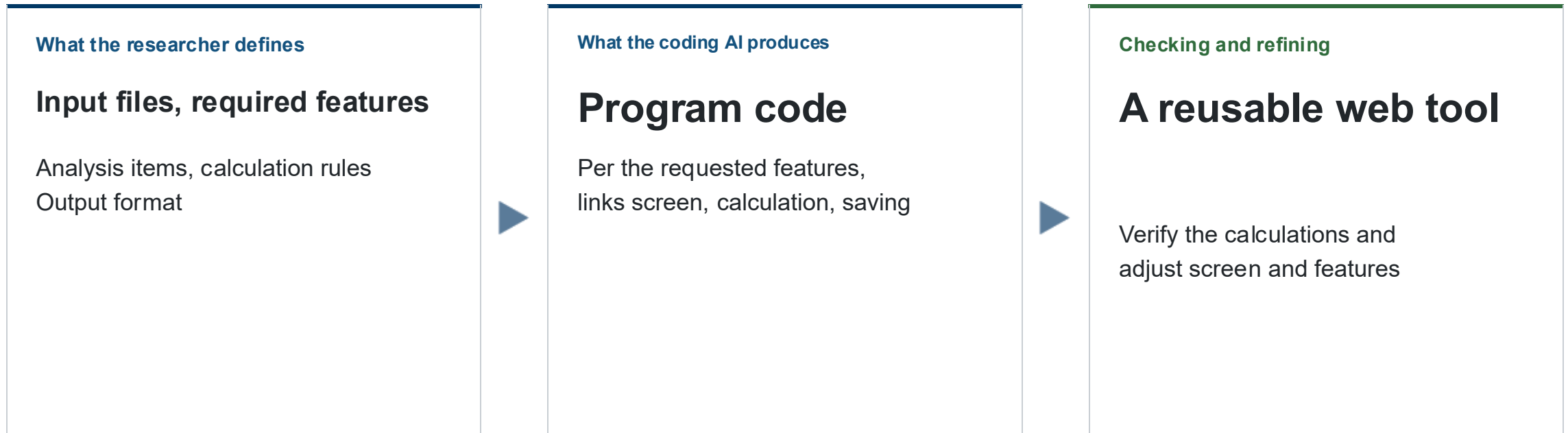
- 01 Data input and processing
- 02 Screen for changing settings
- 03 Result review and saving

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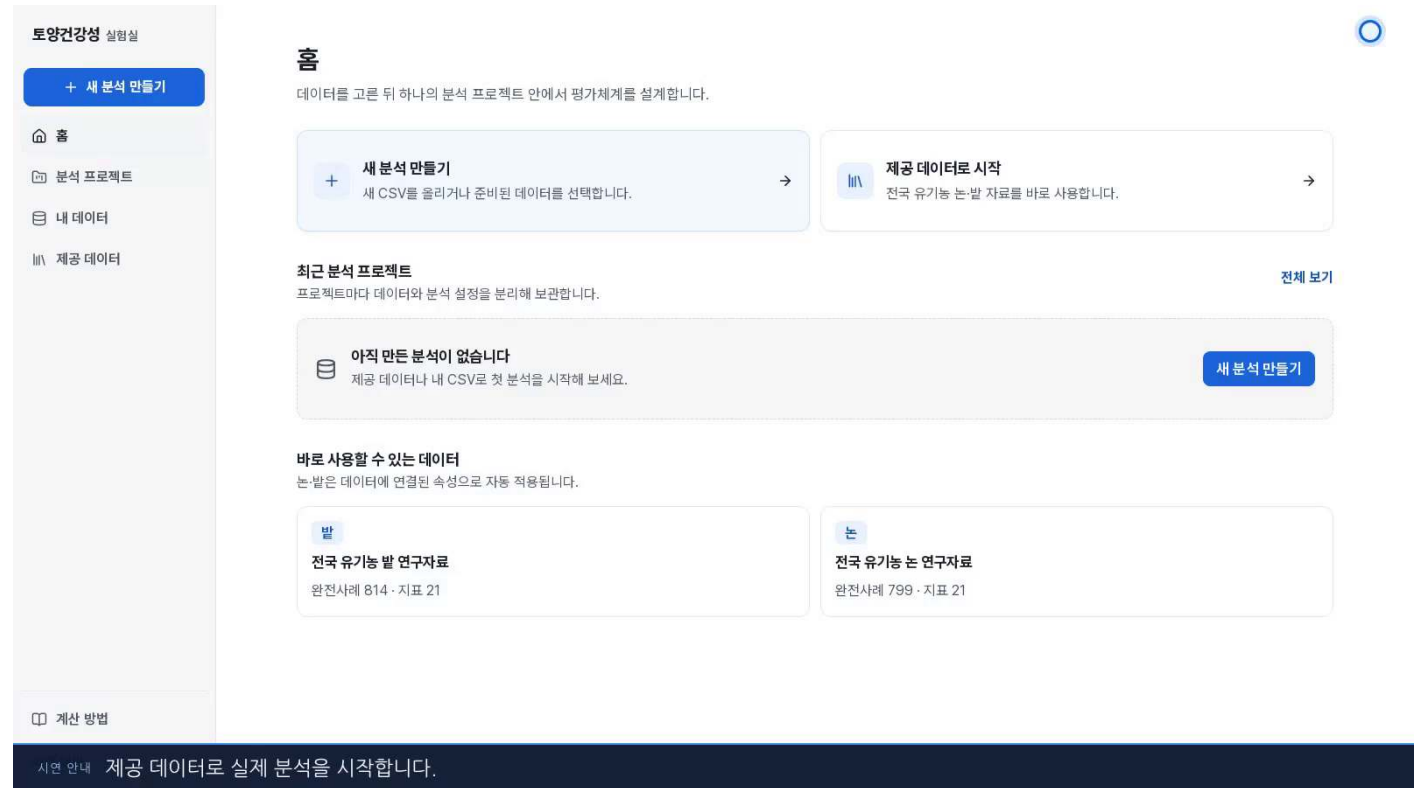
Building and using a web tool

When building



An AI-built web tool in actual use

- 01 Select data, pre-check
- 02 Check distribution, preprocess
- 03 Select and apply assessment items
- 04 Calculate with chosen criteria
- 05 Change settings, compare results
- 06 Review and save results
- 07 Check literature sources



Web dashboard demo · original video 3 min 30 s

04

Large-scale data analysis and synthesis reporting

Re-run with different settings, compile into a report

- 01 Merge multiple Excel files, run analysis
- 02 Generate summary tables, graphs, interpretation
- 03 Compare results under different methods

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Merging 68 Excel files into one synthesis report

Raw records

3.22M record

About 3.21M records used

Unique parcels

1.33M parcels

Parcel count differs from test count

Topics

14 parcels

7 basic + 7 advanced

Outputs

196 tables

25 figures · analysis code (SQL)

Example prompt

Explore the data and write a comprehensive analysis report. Analyse regional differences and relationships between indicators academically, but add plain-language explanations.

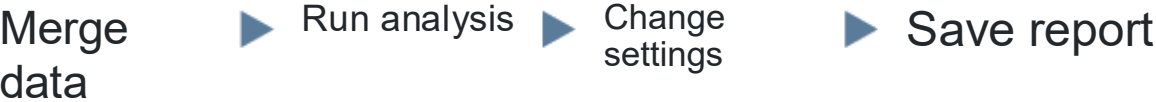
Work extended through follow-up prompts

- Data summary; comparison by type and region
- Yearly trends and relationships between indicators

Raw data: includes some records from 2006-2025

Yearly analysis: 2006-2024

Some indicators (EC, CEC): through 2023



Numeric summaries and written interpretation

Organic matter	g/kg	pH	1:5	Available P2O5	mg/kg
Median	23	Median	6.1	Median	131
Mean	26.96	Mean	6.18	Mean	289.11
Q1-Q3	16.8-31	Q1-Q3	5.6-6.7	Q1-Q3	57-380

Example

Mean available P2O5 is about 2.2× the median

Example interpretation

Distribution summarised by median and middle 50% range (Q1-Q3)

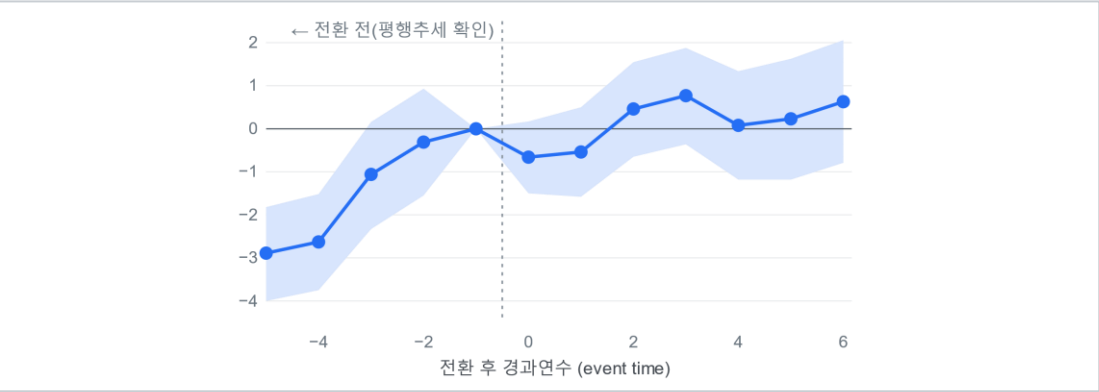
Length and tone adjustable with follow-up prompts

Organic matter, pH n=3,208,035 / available P2O5 n=3,206,139

Comparing results by method and assumption

Example prompt: Change the method and assumptions, and explain why the results differ.

① Adjusting for parcel and year (TWFE)



Y-axis: estimated organic matter effect (g/kg)
X-axis: years relative to conversion (0 = conversion year)

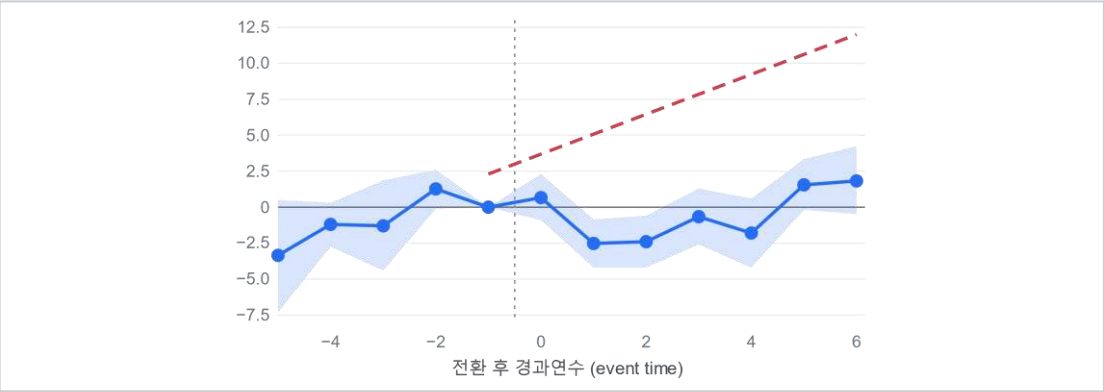
Simple before-after

+2.43 g/kg

After propensity score matching
PSM-DID

+2.78 g/kg

② Comparison by conversion timing (CS) and trend assumption



Blue: CS estimate · Red dashed: pre-conversion trend extended

Adjusted for parcel and year
TWFE

+0.28 g/kg

By conversion timing
Callaway-Sant'Anna

-1.12 g/kg

Estimates vary by method and assumption, so the effect of organic conversion cannot be confirmed

05

Driving existing desktop software directly

Linking analysis and writing across Excel, OriginPro and Word

- 01 Checking Excel data, analysing in OriginPro
- 02 Making a scatter plot, finding literature
- 03 Drafting in Word, fixing formatting

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What screen control and skills do

Computer Use · screen control

Read the screen



Act with mouse and keyboard



Check the updated screen and result

Next demo: Excel → OriginPro → Word

Skill · a bundle of work instructions

Example: a skill for paper tables and figures

- Order for checking input data
- Rules for units and statistical notation
- Output format for tables, figures, references
- Items to check before finishing

Write recurring procedures down for AI to follow

The same skill rules apply to screen control and file work

One workflow across Excel, OriginPro and Word

01 Check raw data in Excel

02 Run statistics in OriginPro

03 Build scatter plot, adjust format

04 Search literature, verify evidence

05 Apply skill, draft in Word

06 Tidy figures, captions, references

07 Check format, save as DOCX

자율 저장

시연_토양원자료.xlsx - 이 PC에 저장됨

검색

파일 홈 삽입 그리기 페이지 레이아웃 수식 데이터 검토 보기 자동화 도움말

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Demo of driving existing software · original video 3 min 05 s

06

Generating slides and applying design

Set font, colour and layout rules, apply to an existing PPT

- 01 Choosing references and design requirements
- 02 Building a template, revising an existing PPT
- 03 Keeping content intact, checking readability

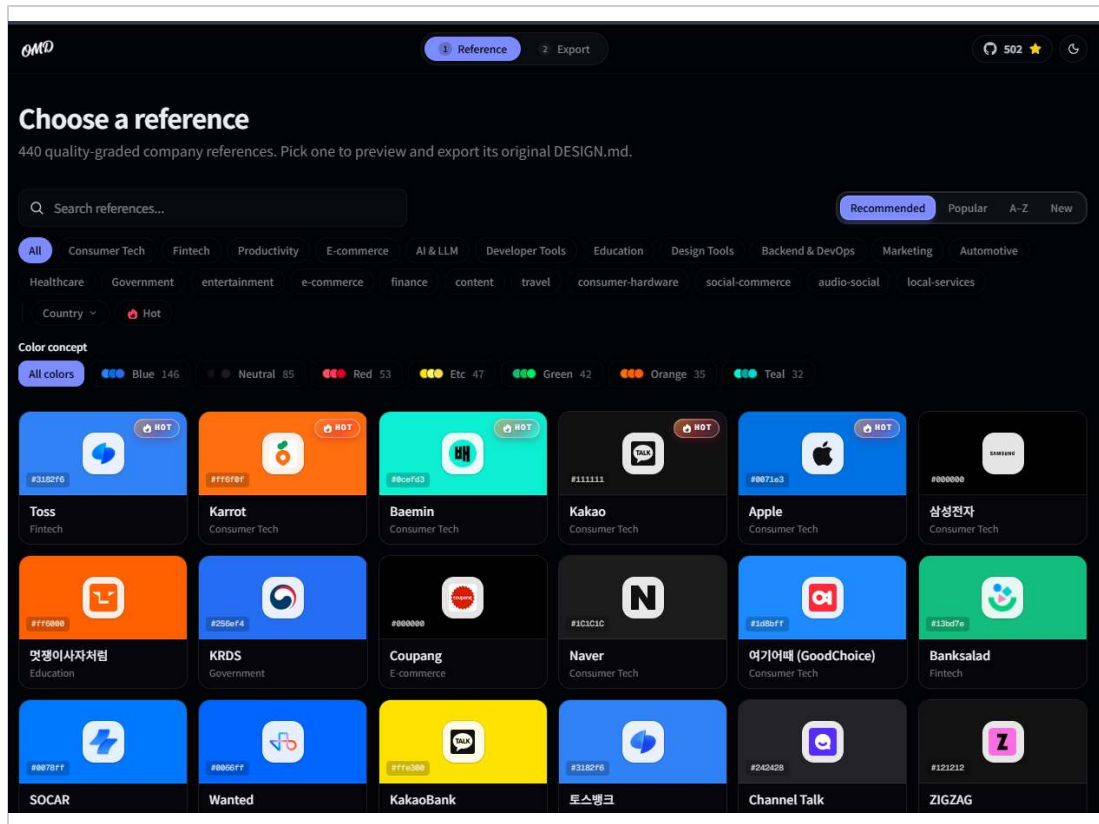
Contents

- 01 How researchers use AI today
- 02 File analysis: tables, figures, reports
- 03 Building reusable analysis tools
- 04 Comparative analysis and synthesis reports
- 05 Driving existing desktop software

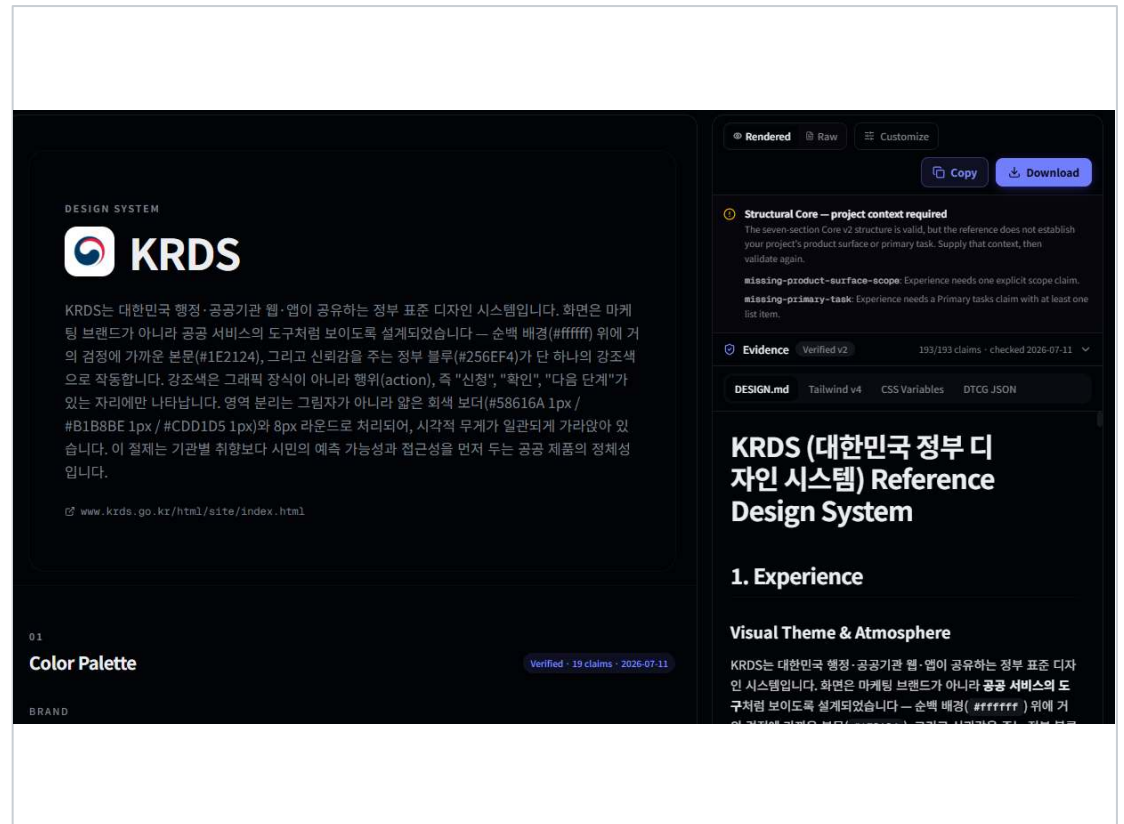
06 Generating slides and applying design

Choosing design references

① List of references

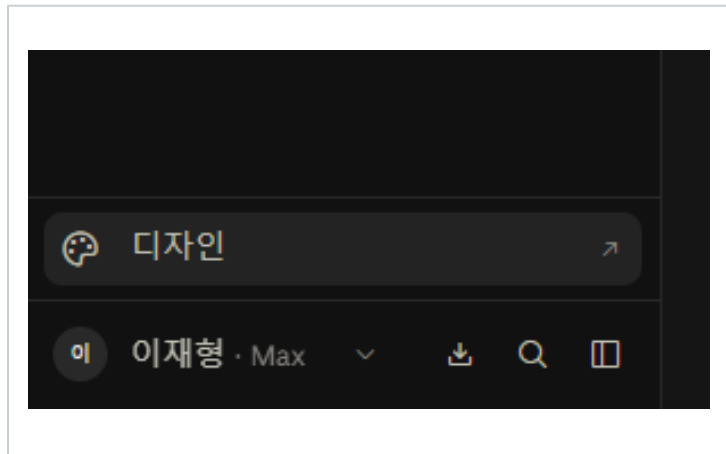


② KRDS rules written up as DESIGN.md

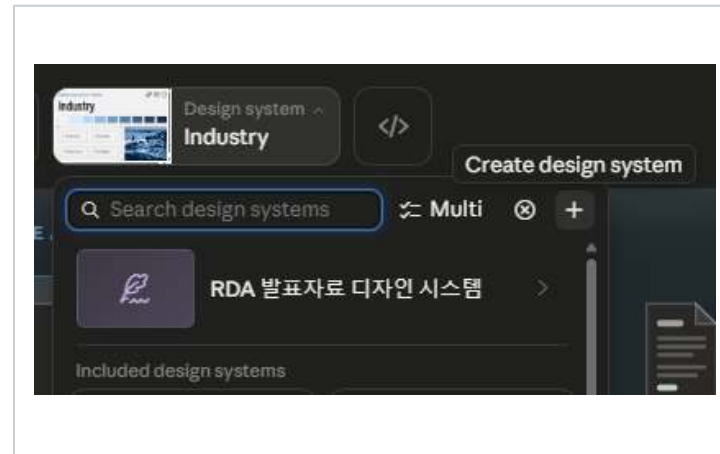


Creating design rules in Claude Design

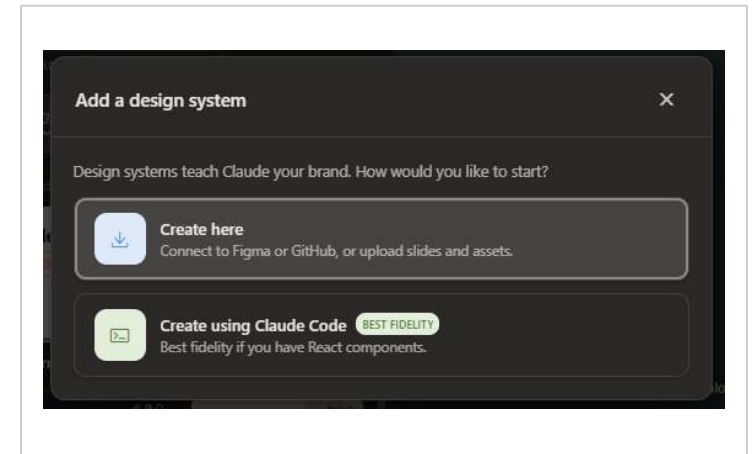
1 Open Design



2 Select Design system



3 Select Create here



Generating design rules from your requirements

Audience and purpose

- RDA research plan and result briefings, training
- 16:9 PowerPoint slides

Visual requirements

- White background, deep blue, green accent
- Priority on Korean readability, tables, graphs, field photos

Attachments

- DESIGN.md and the institute logo

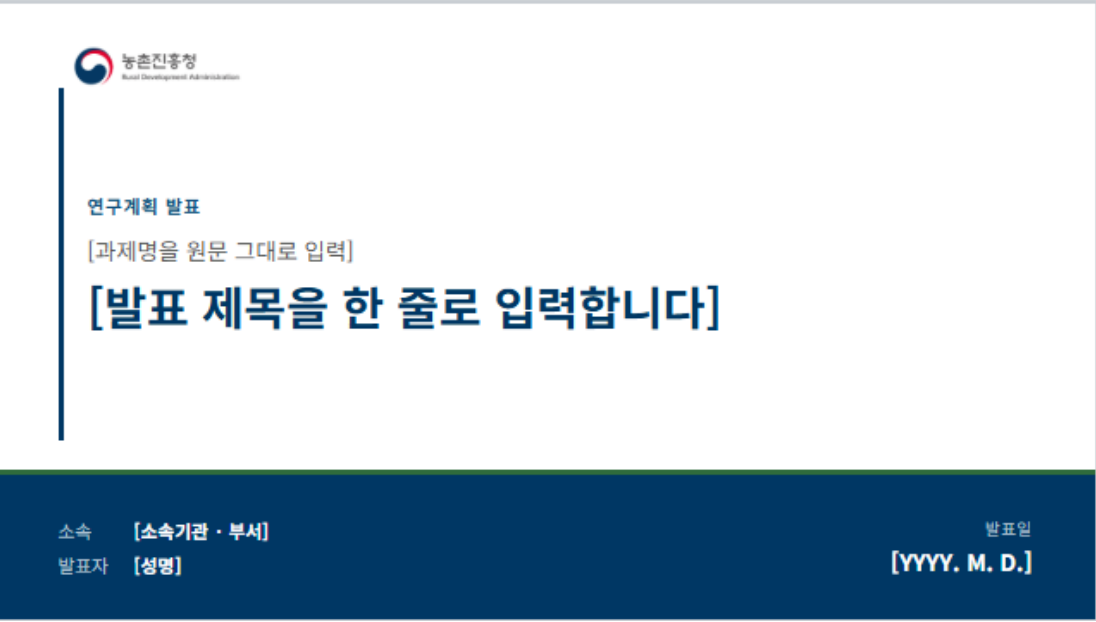
The screenshot shows a web form titled "Set up your design system" with the subtitle "Tell us about your company and attach any design resources you have." The form is divided into several sections:

- Company name and blurb (or name of design system)**: A text area containing Korean text: "첨부한 「DESIGN (4).md」를 참고하여 '농촌진흥청 업무-연구 발표자료용 PPT 디자인 시스템'을 만들어줘. 단순히 PPT 한 편을 예쁘게 만드는 것이 아니라, 앞으로 연구계획, 연구결과, 업무보고, 과제기획, 현장교육 자료를 만들 때 반복해서 적용할 수 있는 디자인 규칙과 재사용 가능한 구성 요소를 만드는 작업이다. 이 결과물은 기관에서 승인한 공식 표준이 아니라, 노초지호청 어르메 화유하기 위한 디자인 시스템에 속합니다".
- Provide examples of your design system and products (all optional)**: A section with the instruction "What works best: code and designs for your design system and your code products." It contains three input fields:
 - Link code from GitHub**: A text box with "https://github.com/owner/repo" and an "Add" button.
 - Link code from your computer**: A text box with "Drag a folder here or browse".
 - Upload a .fig file**: A text box with "Drop .fig here or browse".
- Add fonts, logos and assets**: A text box with "Drag files here or browse".

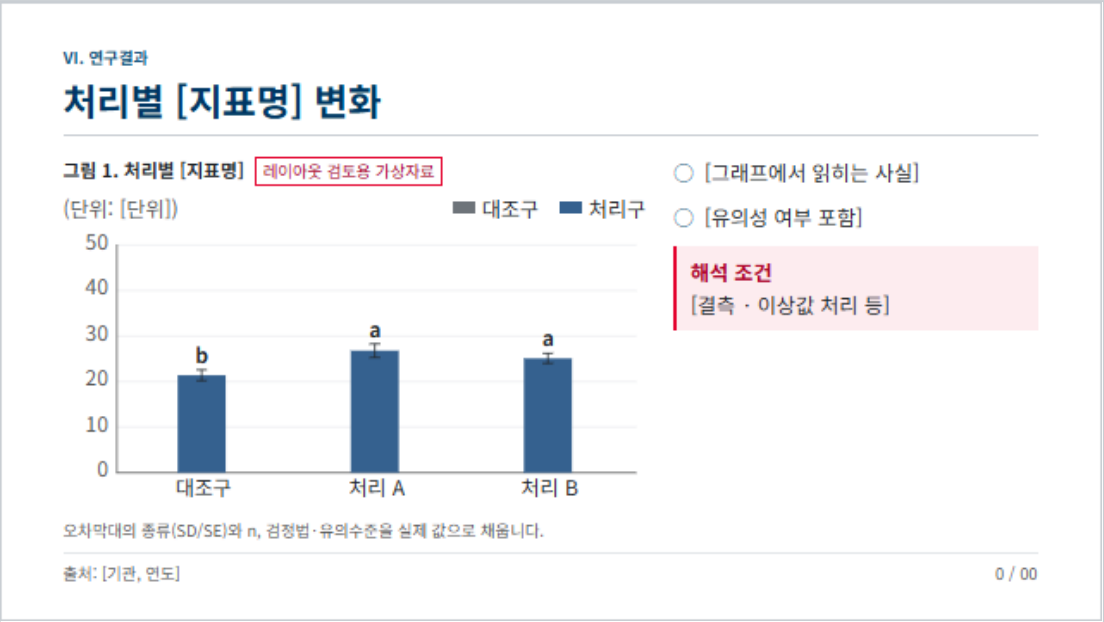
At the bottom, there are two file upload indicators: "DESIGN (4).md" and "농촌진흥청_훈합_좌우2.jpg", each with a close button (X).

Example template generated by AI

① Cover



② Results



Mock data for layout review

Graph values and significance letters are not real research results.

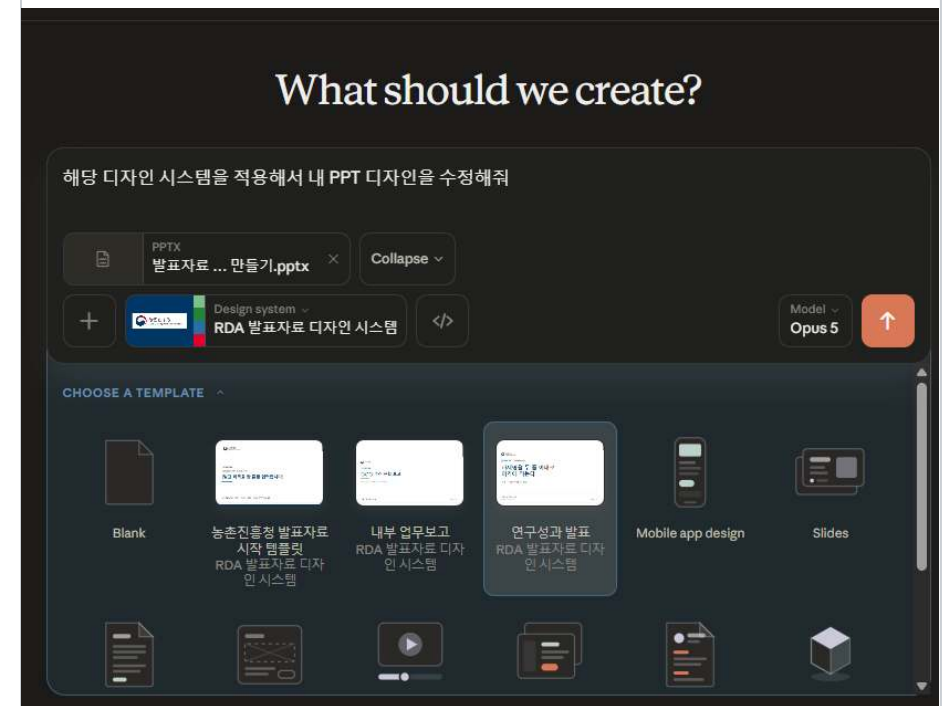
Asking AI to apply design rules to an existing PPT

Input

- Existing PPTX file
- The chosen presentation design system

Refined prompt example

"Apply the design rules.
Keep all values, units, references
and research figures; align titles and margins,
and save as a new PPTX."



The actual deck with the design rules applied

① Cover



농촌진흥청
Rural Development Administration

생성형 AI를 활용한 연구 데이터 분석 · 1 / 40

전문연구원 워크숍

생성형 AI를 활용한 연구 데이터 분석

파일 분석부터 그래프·보고서·발표자료 생성까지

소속 국립농업과학원 재생유기농업과

발표자 이재형

발표일 2026. 9. 18.(금)

② Content slide

PART 4 · 비교 분석과 종합 보고서 작성

수치 요약과 설명문 작성

유기물	g/kg	pH	1:5	유효인산	mg/kg
중앙값	23	중앙값	6.1	중앙값	131
평균	26.96	평균	6.18	평균	289.11
Q1-Q3	16.8-31	Q1-Q3	5.6-6.7	Q1-Q3	57-380

예시

유효인산의 평균은 중앙값의 약 2.2배

설명문 예시

중앙값과 중간 50% 범위 (Q1-Q3)로 분포 요약

추가 요청으로 분량·문체 조정 가능

유기물·pH n=3,208,035 / 유효인산 n=3,206,139

흙토람 토양검정 종합 인사이트 보고서(심화판), 2026. 8. 11. · 정제지표 분포 요약표

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Drafted in Claude Design and Claude Code, then revised in Codex

Research Fellows Workshop

Analyse research data with AI - right through to the slides

Define the target, the calculation conditions and the output format, then check results against the raw data.

Thank you